TCS NQT STRATEGY SERIES

Quantitative Aptitude: Statistics Masterclass

Mean, Median, Mode, Standard Deviation & Variance

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Section 1: Conceptual Clearance & Shortcuts

In the TCS NQT, statistics questions are designed to test your ability to derive properties from data sets quickly, not just plug numbers into formulas. Focus on the relationships between these central tendencies.

1. Central Tendencies

Core Formulas:

- **Mean (*ar{x}*):** Sum of all observations / Total number of observations.
- **Median:** Middle value in a sorted list. If *n* is odd, *((n+1)/2)^{th}* term. If *n* is even, average of *(n/2)^{th}* and *(n/2 + 1)^{th}* terms.
- **Mode:** The observation with the highest frequency.
- **Empirical Relationship:** *ext{Mode} = 3(ext{Median}) - 2(ext{Mean})*. This is a crucial shortcut for TCS questions!

2. Dispersion Metrics

Variance (*\sigma^2*) & Standard Deviation (*\sigma*):

- *ext{Variance } (\sigma^2) = rac{\sum (x_i - ar{x})^2}{n}*
- *ext{Standard Deviation } (\sigma) = \sqrt{ ext{Variance}}*
- **Shortcut:** *ext{Variance} = (ext{Mean of squares}) - (ext{Square of mean})*.

Section 2: TCS Traps

TRAP 1: THE "ADDED OBSERVATION" MEAN SHIFT

TCS loves giving you an initial mean and then adding one value, asking for the new mean or the value of that added observation.
Shortcut: $ ext{New Value} = ext{Old Mean} + (n+1) imes (ext{New Mean} - ext{Old Mean})$. Do not calculate long sums!

TRAP 2: DIGIT INTERCHANGE CONFUSION

A common trap is replacing a number (e.g., 28) with its interchanged version (82) in a mean calculation. **Correction:** The change in sum is simply the difference between the interchanged values. $ ext{New Sum} = ext{Old Sum} + (ext{Wrong Number} - ext{Correct Number})$.

Section 3: 10 High-Priority Practice Questions